

Regression, pt. 3

CSU Chico, Math 351

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Covariance

Given a pair of random variables, how does one vary with respect to the other? If X increases, for example, does Y increase as well? And if so, how strong is the dependence between the two? For this, we define covariance:

Given random variables X and Y with finite variances, define the *covariance* of X and Y , written $\text{Cov}(X, Y)$, as

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

If X and Y are independent, then $\text{Cov}(X, Y) = 0$.

Suppose X and Y are random variables with finite variances, and a and b are constants. Then

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

Correlation Coefficient

The covariance of X and Y necessarily reflects the units of both variables, which can make interpretation difficult. It helps to have a dimensionless measure of dependence so that one xy relationship can be compared to another. Dividing the covariance by $\sigma_X\sigma_Y$ accomplishes this and bounds the value between -1 and 1.

Definition: Let X and Y be any two random variables. The correlation coefficient of X and Y , denoted $\rho(X, Y)$ is given by

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X\sigma_Y} = \text{Cov}(X^*, Y^*)$$

where $X^* = (X - \mu_X)/\sigma_X$ and $Y^* = (Y - \mu_Y)/\sigma_Y$.

Estimating R

Suppose the correlation coefficient between X and Y is unknown, but we have some relevant information about its value in the form of n measurements $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$. How can we use those data to estimate $\rho(X, Y)$?

We define the *sample correlation coefficient* to be the ratio

$$R = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}}{\sqrt{\frac{1}{n} \left(\sum_{i=1}^n X_i - \bar{X} \right)^2} \sqrt{\frac{1}{n} \sum_{i=1}^n \left(Y_i - \bar{Y} \right)^2}}$$

or equivalently:

$$R = \frac{n \sum_{i=1}^n X_i Y_i - \left(\sum_{i=1}^n X_i \right) \left(\sum_{i=1}^n Y_i \right)}{\sqrt{n \left(\sum_{i=1}^n X_i^2 \right) - \left(\sum_{i=1}^n X_i \right)^2} \sqrt{n \left(\sum_{i=1}^n Y_i^2 \right) - \left(\sum_{i=1}^n Y_i \right)^2}}$$

Interpreting R

Another way to judge the quality of a regression model focuses on the square of R . The idea with R^2 is that it measures the proportion of the variation in the response variable that is explained by the regression model or x . We can rewrite the sum of the residuals squared as

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^n (y_i - \bar{y})^2 - \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2$$

Interpreting R

Additionally, given the maximum likelihood estimate for slope, $\hat{\beta}_1$:

$$r = \hat{\beta}_1 \frac{\sqrt{n * \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2}}{\sqrt{n * \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2}}$$

This, plus the fact that $\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2 / n$, means we can write

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^n (y_i - \bar{y})^2 - r^2 \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x})^2$$

which reduces to

$$r^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

R^2 Properties

R^2 has some nice properties. Notice that

1. $(y_i - \bar{y})^2$ represents the total variability in the dependent variable.
2. $\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$ represents the variation in the y_i 's not explained by the regression model.
3. $(y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$ represents the variation in the y_i 's that is explained by the linear regression with x .

Therefore, the interpretation of r^2 is *the proportion of the total variation in the y_i 's that can be attributed to the linear relationship with x .*

For instance, $r = 0.70$ suggests 49% of the variation in Y is explained by the linear regression with X (the other 51% is associated with other factors).

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Example 1

A study on SAT scores monitored one testing period's average SAT scores (Y), by state, as a function of participation rate (X), where the SAT score is the sum of the Critical Reading, Math, and Writing subtest scores (out of 2400 total). To observe the overall relationship between the scores and participation rates, calculate the sample correlation r based on the following sums:

$$\sum_{i=1}^n X_i = 1,891; \quad \sum_{i=1}^n Y_i = 81,396$$

$$\sum_{i=1}^n X_i^2 = 114,983; \quad \sum_{i=1}^n Y_i^2 = 130,597,738$$

$$\sum_{i=1}^n X_i Y_i = 2,863,056$$

Example 1

Taking the summed quantities given and the formula for r , we can find:

$$r = \frac{51(2,863,056) - (1,891)(81,396)}{\sqrt{51(114,983) - (1891)^2} \sqrt{51(130,597,738) - (81,396)^2}} = -0.881$$

This implies a strong negative linear association.

$r^2 = (-0.881)^2 = 0.776$, and we can say 77.6% of the variability in SAT scores from state to state can be attributed to the linear relationship between test scores and participation rates.

Exercise 1

The dataset “corn.csv” (found in the CodiMD notes and the table below) shows U.S. corn supplies in millions of bushels and corn prices in dollars per bushel for 1999 to 2008. Identify all relevant sums and calculate the sample correlation coefficient, r .

Year	Supply(x)	Price(y)
1999	11.2	1.7
2000	11.8	1.8
2001	11.5	2
2002	10.6	2.4
2003	11.2	2.5
2004	12.8	2.1
2005	13.1	2
2006	12.6	3
2007	14.5	4.2
2008	13.8	4.1

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Adapted from notes by Dr. Gray & Dr. Roualdes.